

TEZPUR UNIVERSITY
Test III, Spring 2022
CO314 System Software and Compiler Design

Time 45 min

Marks 25

All questions are compulsory.

The figures in the right-hand margin indicate marks for the individual question.

1. Consider the following Context Free Grammar:

$S \rightarrow a B \mid D e$

$B \rightarrow b D \mid D$

$D \rightarrow d D \mid e$

- (a) What would be the LEADING and TRAILING sets? 6
- (b) Write the augmented for LR(k) parsing for the above grammar. 1
- (c) Write the first LR(0) item set for the above grammar. 2

2. Rewrite the following grammar to make it suitable for LL(1) parsing. 2

$E \rightarrow E + T \mid T$

$T \rightarrow T * F \mid F$

$F \rightarrow n \mid (E)$

3. Represent the following expression in the form of a syntax tree assuming common precedence and associativity. 2

$a * b + c + b$

4. Indicate in which memory segment will each of the data objects of the following C program segment be allotted memory. 5

int x;

void f1(char b)

{

 short

 a;

```

static int      s;
a = 2 * b;
if( a < 20 ){
    short  d;
    d = a++;
}
return;
}

```

5. Identify and categorise the error in the following C program segment. 2

```

int f(){
    int    a, b;
    float  f, *p = &f;
    scanf( "%d %d", &a, &b );
    f = a / p;
    return( b );
}

```

6. Consider the following three-address code sequence.

```

T1 = a + b
x = T1
T2 = a + b
T3 = T2 * c
y = T3
if x <= 10 goto (7)

```

- (a) Draw a DAG for the above 3AC sequence. 3
 (b) Prepare an optimised 3AC sequence for the above. 2

T1 = T2

TEZPUR UNIVERSITY

End Semester Examination, Spring 2022

CO 314 System Software and Compiler Design

Time : 3 hours

Marks : 60

All questions are compulsory. The figures in the right-hand margin indicate marks for the individual question.

Submit the answers to question 13 within first 15 minutes

1. (a) Write regular expressions for three different types of tokens of an assembly language program. 3

(b) How do macros facilitate higher execution efficiency compared to function calls? 2

2. Write a context free grammar G for the *if-construct* of C language (with simple constituents for the *condition* and *statements*). Explain why the construct is ambiguous. 2 + 3

3. (a) Write suitable routines for recursive descent parsing of *while-construct* of a high level language. 4

OR

(b) What are the conditions that implies that a grammar is not LL(1)? 4

4. Though an LALR parser merges certain LR(1) item sets computed for a CLR parser, it does not encounter any additional *shift-reduce* conflict compared to the latter. Why? 3

5. (a) Why does a compiler produce *intermediate code* instead of directly producing the target code? 3

(b) Represent the following expression in postfix form- 2

$$x = a + b * c$$

6. Write semantic actions to produce three address code for input corresponding to the grammar rule-

$$E \rightarrow E + T$$
 2
7. When and where are *activation records* created? 2
8. (a) Explain with examples the complex phenomena of scope, lifetime and binding of identifiers and data objects in a block structured language and the impact on implementation of symbol table. 4 + 2
- (b) Why may a binary search tree be used for a symbol table instead of a sorted array that can facilitate binary search? 2
9. Write two desired characteristics of error reporting by a compiler. 2
10. Consider the following three-address code sequence, S -
- | | |
|--------------------------|--------------------------------|
| (1) if $n < 10$ goto (3) | (6) $p = T2$ |
| (2) goto (5) | (7) $T3 = p * 4$ |
| (3) $i = j$ | (8) $T4 = \text{addr}(s)[T3]$ |
| (4) goto (11) | (9) $T5 = p * 4$ |
| (5) $T2 = n + 1$ | (10) $\text{addr}(x)[T5] = T4$ |
- (a) Identify the basic blocks. 2
- (b) Draw the DAG for the code portion from (5)-(10). 4
11. (a) Suppose for a particular step in a program, it is required to have the value of a data object x in a CPU register, and that all the registers are already used for other data objects. How can the address descriptors and register descriptors be used and updated for this purpose? 4
- OR
- (b) How is static linking different from dynamic linking? 4
12. What is program relocation? What information should a compiler produce for relocation? 2 + 2

TEZPUR UNIVERSITY
Mid Term Examination, Spring 2022
CO314 System Software and Compiler Design

Marks : 40

Time : 2 hours

The figures in the right-hand margin indicate marks for the individual question.

Answers should be concise and entire answer to a question should be together. Give examples and diagrams wherever appropriate. State assumptions wherever made.

-
1. Since the assembler writes the output machine language program into a file instead of putting it into memory, how is the address specified by the assembler directive *START <addr>* used by the assembler? 1
 2. (a) Write a *flex* input file for creating a lexical analyser to recognize the keywords *MACRO* and *MIF* and macro parameters that are identifiers preceded by the *\$* symbol. 4
 - 4 (b) What are the different situations in a program where subroutines and macros are suitable for use? Why? 3
 3. Give an example of a macro definition and describe how a macro preprocessor will store information about that macro in the major preprocessor data structures. 2 + 4
 - ⑦ How is *conditional expansion* of a macro useful? Explain with an example. 4
 - ⑤ What are some procedural and data abstractions found in high level programming languages? 3

6. (a) Compared to assembly languages do compilers enhance programming ease and translation ease? 2

OR

- (b) How is a compiler different from an interpreter? 2

7. Show with an example how a single high level language statement may be translated to multiple machine (or assembly) language statements. 2

8. (a) Show with an example the significance of proper sequence of machine language steps produced for a high level language statement. 2

- (b) What are associativity and precedence of operators? What are the precedence relations between operators? 3

9. (a) For a set of arithmetic operators, describe how you will represent the precedence relations in suitable data structures in a C program. 4

OR

- (b) Describe how a stack is used in evaluation of arithmetic expressions. 4

10. Write a context free grammar to describe the syntax of *while*-construct of C language. 3